M. SUDALAI

Aspiring Analytics Engineer | Data Analytics & Machine Learning

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PROFESSIONAL SUMMARY

B.Tech student in **Artificial Intelligence and Data Science** with hands-on experience in **Python, SQL, and Machine Learning** through academic projects and a data analytics internship. Currently building advanced skills in **data warehousing, dbt, and pipeline orchestration** to pursue a career as an **Analytics Engineer**. Strong foundation in data analysis, model comparison, and problem-solving.

SKILLS

- **Programming & Querying:** Python, SQL
- **Machine Learning & Data Science:** Scikit-learn, XGBoost, TensorFlow, Keras, RDKit, Pandas, NumPy
- **Visualization:** Matplotlib, Seaborn
- **Currently Learning:** dbt, Data Warehousing, Apache Airflow, Advanced SQL *(as part of an analytics engineering roadmap)*

EDUCATION

Higher Secondary Education — **2021 – 2023**
Govt. Boys Hr. Sec. School, Kayanallur, Tenkasi | Score: **80%**

B.Tech, Artificial Intelligence and Data Science — **2023 – 2027 (Expected)**
Easa College of Engineering and Technology, Coimbatore | CGPA: **8.1**

PROJECTS

TNBC Drug Discovery & Comparative Study
Repo: https://github.com/msudalaivasu-droid/Drug-discovery-and-comparative-study-.-

- Predicting how well a drug is absorbed (**bioavailability**) is a key step in screening candidates for **triple-negative breast cancer (TNBC)**, and the initial baseline model was not accurate enough to rely on.
- Build a more accurate, reliable prediction pipeline using a **1,052-compound** lab-tested dataset , make it ready to scale to **10,000+ compounds** for real-world drug screening.
- Engineered **2,757 molecular features** (30 physicochemical properties + 2,727 fingerprints from RDKit's Morgan, MACCS, and Avalon methods). Then compared four approaches — a **residual deep neural network, XGBoost, a stacked cross-validated ensemble, and a weighted ensemble** — and tuned the ensemble weights with **SLSQP**. Built using **Python, TensorFlow/Keras, Scikit-learn, XGBoost, and RDKit**.
- The best model reached **80.39% accuracy (R^2 = 0.80)**, a **10.3% improvement** over the baseline, giving a dependable pipeline for screening TNBC drug candidates.

CERTIFICATIONS

- **Data Analytics Internship** — Ideal Formatic Institute of Technology (June–July 2025, 17 days)

LANGUAGES

- **Tamil**
- **English**